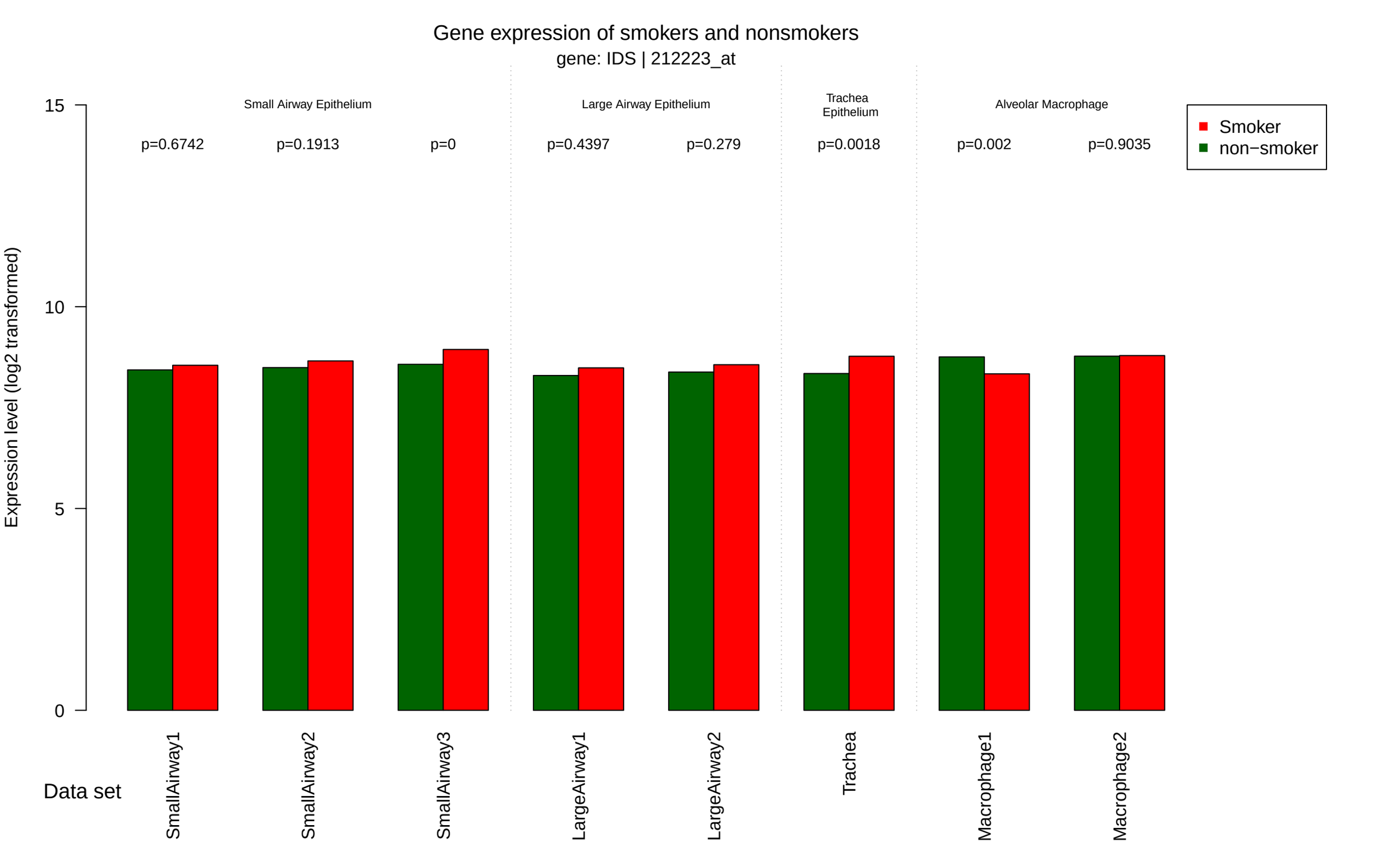
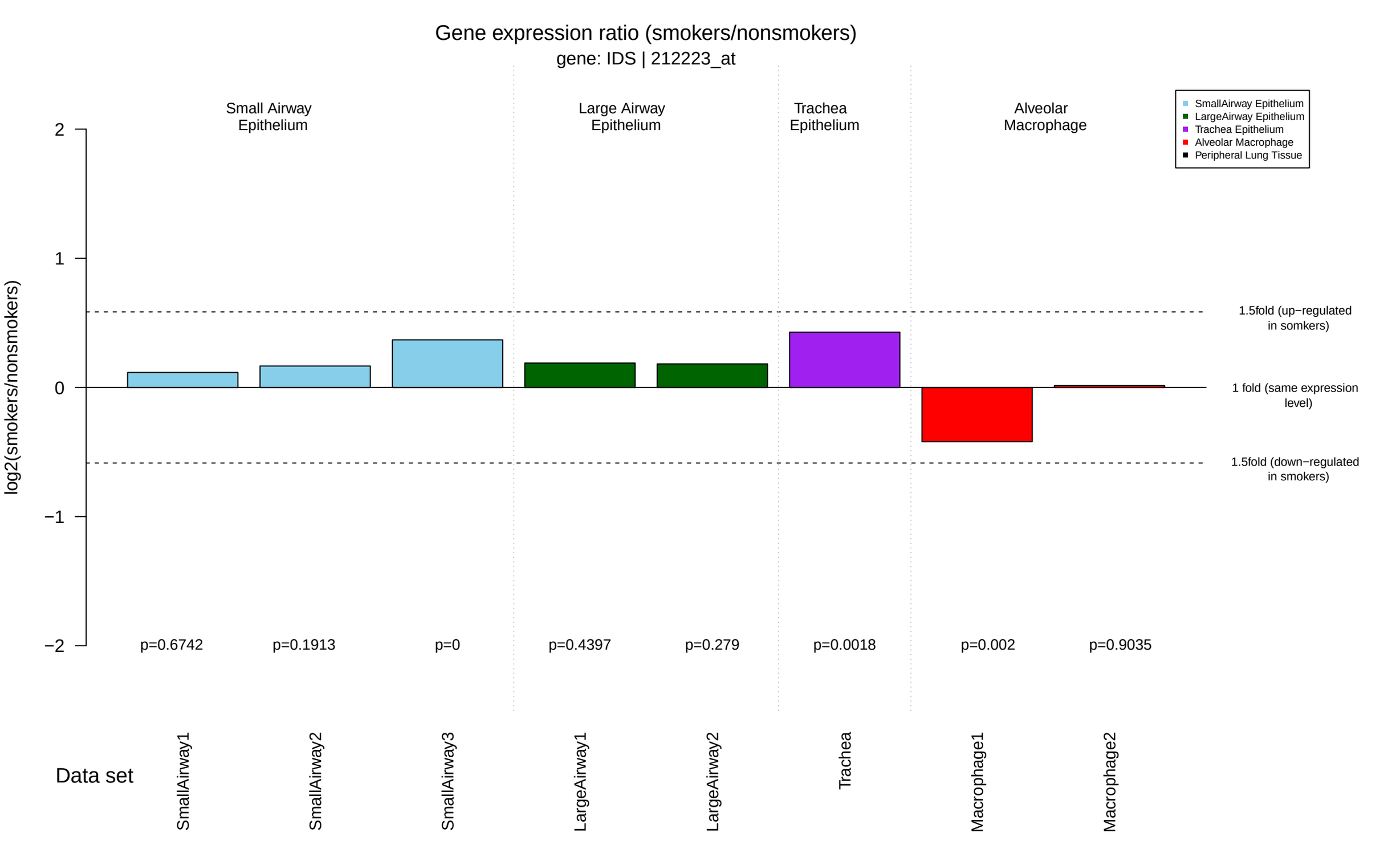
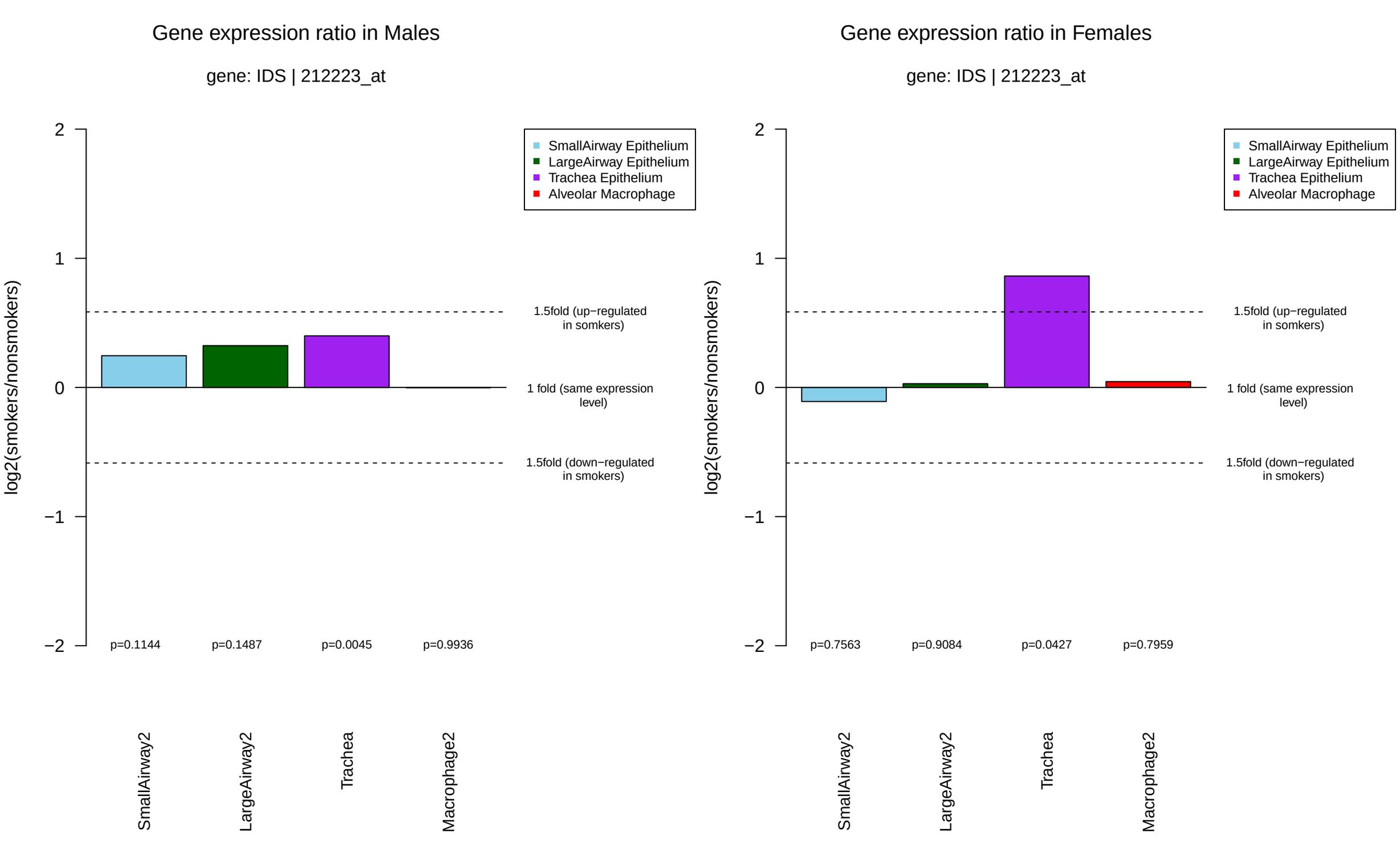
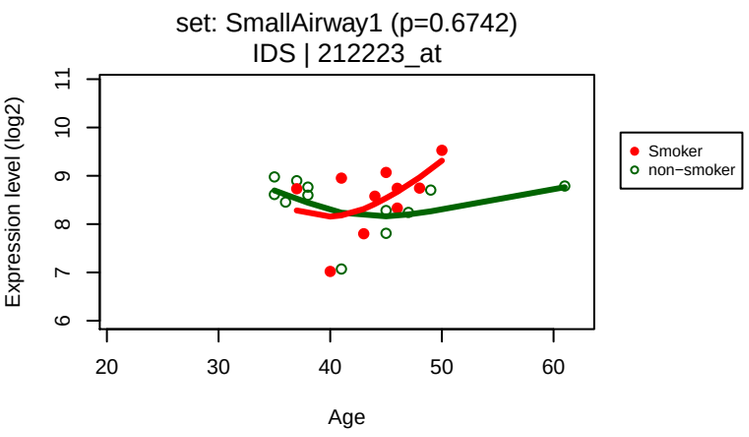
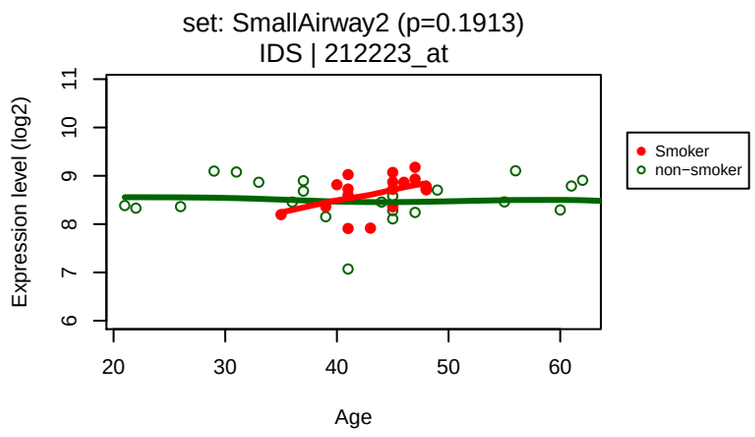
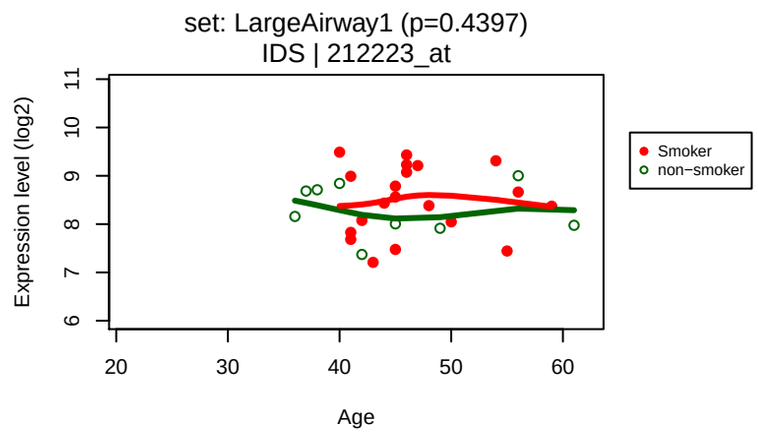

Gene expression of smokers and nonsmokers
gene: IDS | 212223_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6742
p=0.1913
p=0
p=0.4397
p=0.279
p=0.0018
p=0.002
p=0.9035
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

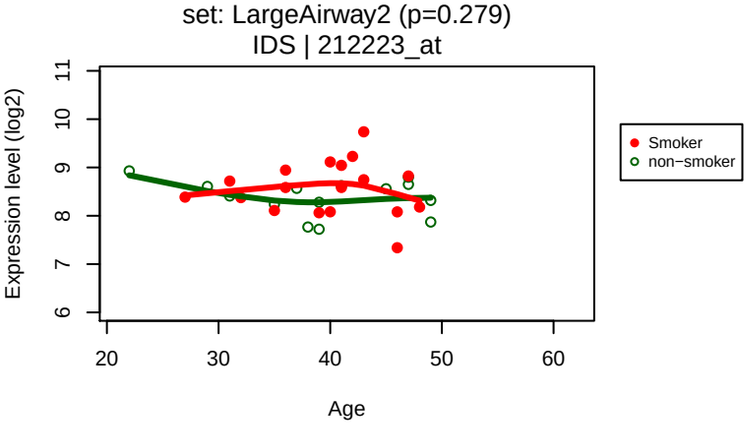

Gene expression ratio (smokers/nonsmokers)
gene: IDS | 212223_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6742
p=0.1913
p=0
p=0.4397
p=0.279
p=0.0018
p=0.002
p=0.9035
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

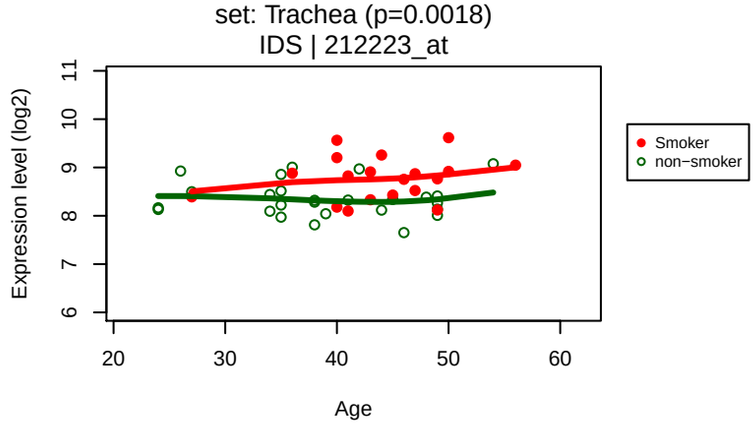

Gene expression ratio in Males
gene: IDS | 212223_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1144
p=0.1487
p=0.0045
p=0.9936
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IDS | 212223_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7563
p=0.9084
p=0.0427
p=0.7959
SmallAirway2
LargeAirway2
Trachea
Macrophage2

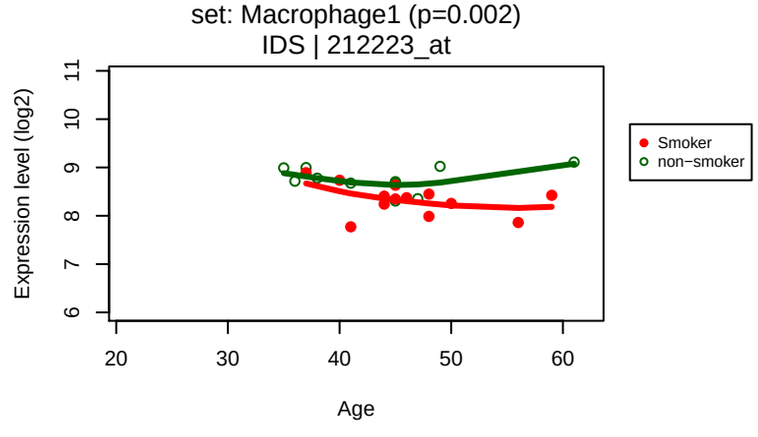

set: SmallAirway1 (p=0.6742)
IDS | 212223_at
Expression level (log2)
Age
Smoker
non−smoker

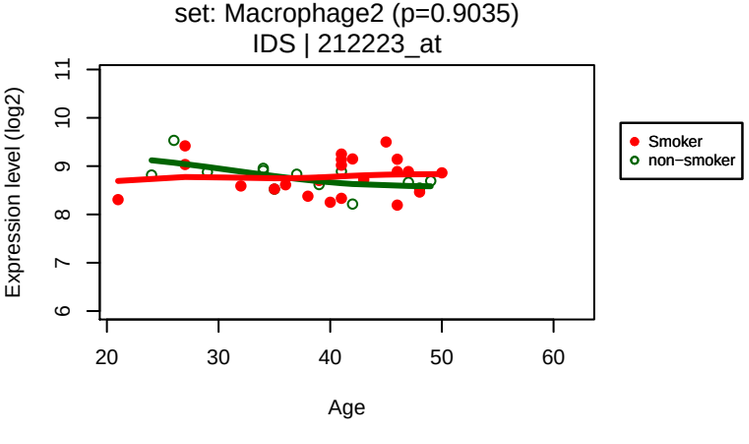

set: SmallAirway2 (p=0.1913)
IDS | 212223_at
Expression level (log2)
Age
Smoker
non−smoker

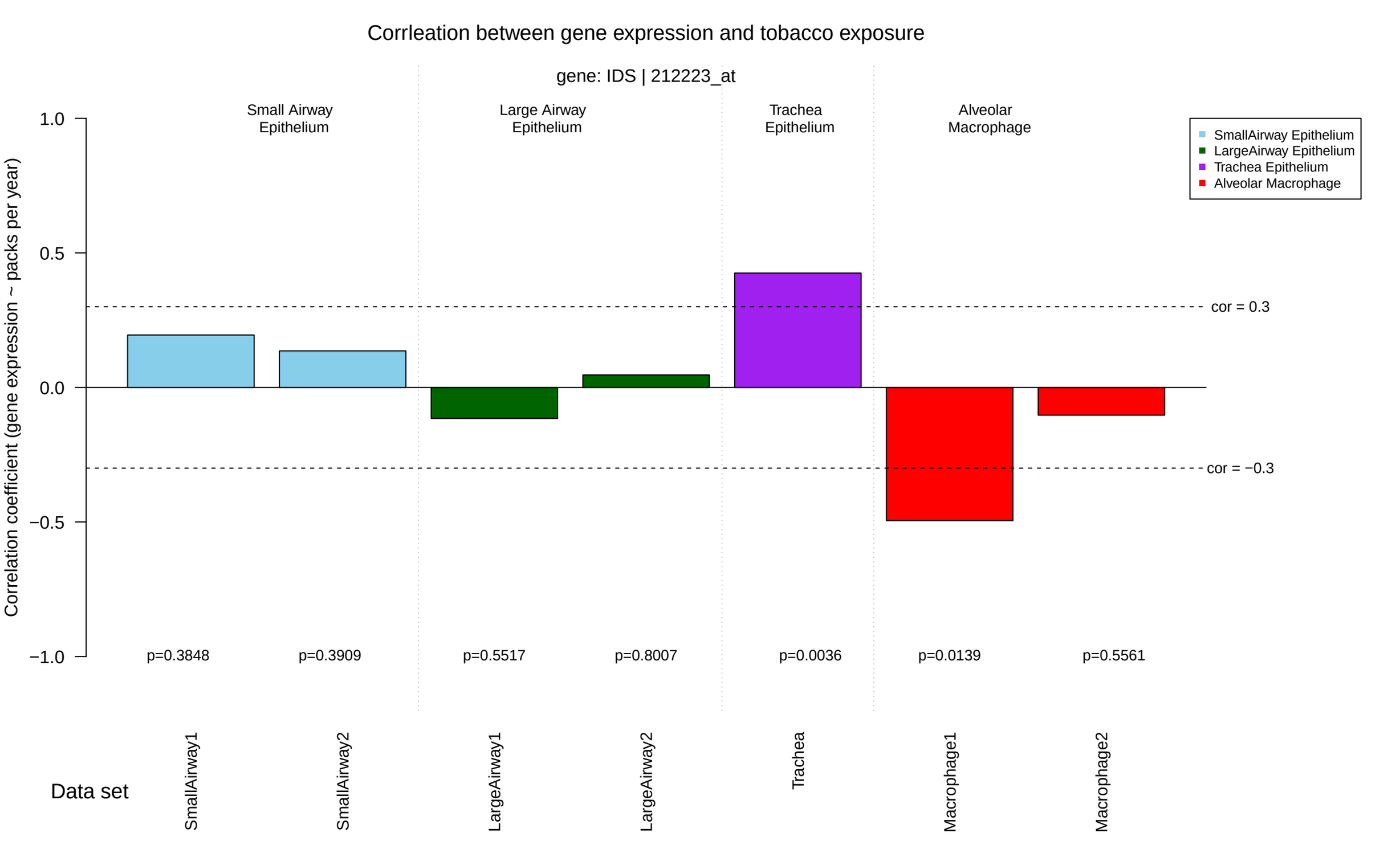

set: LargeAirway1 (p=0.4397)
IDS | 212223_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.279)
IDS | 212223_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0018)
IDS | 212223_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.002)
IDS | 212223_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9035)
IDS | 212223_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IDS | 212223_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3848
p=0.3909
p=0.5517
p=0.8007
p=0.0036
p=0.0139
p=0.5561
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2